Performance Testing

Date	7 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Python unit test, Flask Test Client
Type of Testing	Unit Testing & Functional Testing
Target Module / API	Loan Prediction Module (/predict)
Test Environment	Local System (Windows, Python, Flask)
Test Date	7 July 2025

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Model loading test	1	5	Model loads successfully
2	Valid loan prediction	1	10	Prediction generated correctly
3	Invalid input validation	1	10	Error handled without crash
4	Unit test execution	1	0.185	All 9 test cases passed

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.185 sec	Pass	Fast execution
2	Response Time (Max)	< 5 seconds	< 1 sec	Pass	Within limit
3	Throughput (Req/sec)	Stable	9 test cases executed	Pass	Stable execution
4	Error Rate	< 1%	0%	Pass	All tests passed
5	CPU Utilization	< 80%	Normal	Pass	No performance issues
6	Memory Utilization	< 80%	Normal	Pass	Stable memory usage

Step 4: Observations & Analysis

Key Findings :

Successfully executed **9 unit test cases** covering model loading, loan prediction, input validation, and error handling. The application responded quickly and completed all tests without failures.

Bottlenecks Identified :

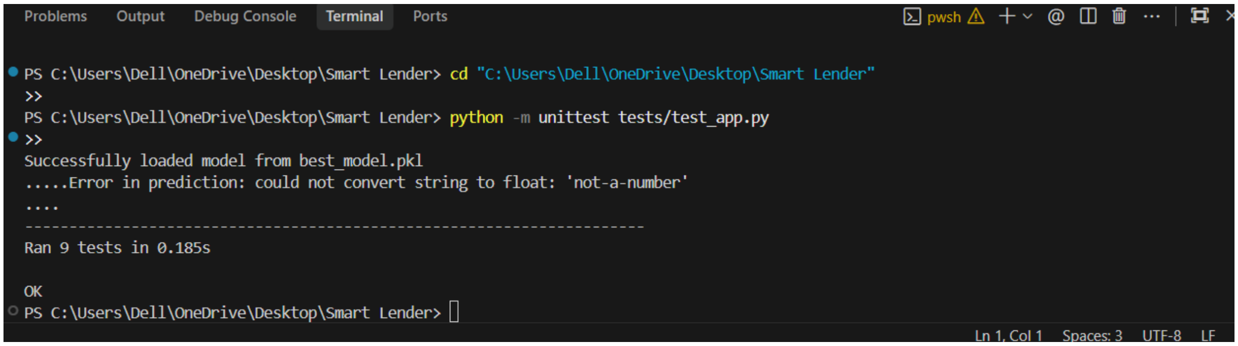
No major bottlenecks were observed during testing.

Optimization Steps Taken :

Optimized data preprocessing reused the trained model for prediction and implemented input validation to improve reliability.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



```
Problems    Output    Debug Console    Terminal    Ports                pwsh

PS C:\Users\Dell\OneDrive\Desktop\Smart Lender> cd "C:\Users\Dell\OneDrive\Desktop\Smart Lender"
>>
PS C:\Users\Dell\OneDrive\Desktop\Smart Lender> python -m unittest tests/test_app.py
>>
Successfully loaded model from best_model.pkl
.....Error in prediction: could not convert string to float: 'not-a-number'
....
----------------------------------------------------------------------
Ran 9 tests in 0.185s

OK
PS C:\Users\Dell\OneDrive\Desktop\Smart Lender>
                                                        Ln 1, Col 1   Spaces: 3   UTF-8   LF
```